American Computer Science League

2025-2026 ● Contest 1: Short Problems ● Junior Division

1. Computer Number Systems

Convert 52326_{16} to octal.

A. 336677
B. 1221446
C. 2443112
D. 1222045
E. 1622045

2. Computer Number Systems

What is the average of the following two numbers in base 10?

36_8 74_8

A. 36
B. 37
C. 45
D. 112
E. 132

3. Recursive Functions

Find $f(6)$ given:

$$f(x) = \begin{cases} 2 & if\ x = 1 \\ 2 \cdot f(x-1) + 1 & if\ x > 1 \end{cases}$$

A. 5
B. 11
C. 23
D. 95
E. 191

4. Recursive Functions

Find $f(16)$ given:

$$f(x) = \begin{cases} f(x-3) + x & if\ x > 10 \\ 2 \cdot f(x-2) & if\ 8 \le x \le 10 \\ 2x + 3 & if\ x < 8 \end{cases}$$

A. 73
B. 15
C. 60
D. 30
E. 89

5. What Does This Program Do? (Branching)

Which of the following input values will print "wrong" when executed?

```
input a, b
a = a ^ 2
b = b ^ 2
c = a + b
d = sqrt(c)
if d == int(d) then
     output a, b, c
else
     output "wrong"
end if
```

A. a=3, b=4
B. a=5, b=12
C. a=5, b=10
D. a=9, b=12
E. a=7, b=24

6. What Does This Program Do? (Branching)

Run the following program with each number from 1 to 25, inclusive. How many of them are output?

```
input num
if num % 3 == 0 && num % 4 == 0 then
     output num
end if
if num % 5 == 0 || num % 7 == 0 then
     output num
end if
```

A. 11
B. 10
C. 9
D. 8
E. 7

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1. 计算机数字系统

将 52326_{16} 转换为 8 **进制。**

A. 336677
B. 1221446
C. 2443112
D. 1222045
E. 1622045

2. 计算机数字系统

以下两个数在十进制下的平均值是多少？

36_8 74_8

A. 36
B. 37
C. 45
D. 112
E. 132

3. 递归函数

根据**下列表达式，求** $f(6)$ **：**

$$f(x) = \begin{cases} 2 & if\ x = 1 \\ 2 \cdot f(x-1) + 1 & if\ x > 1 \end{cases}$$

A. 5
B. 11
C. 23
D. 95
E. 191

4. 递归函数

根据**下列表达式，求** $f(16)$ **：**

$$f(x) = \begin{cases} f(x-3) + x & if\ x > 10 \\ 2 \cdot f(x-2) & if\ 8 \le x \le 10 \\ 2x + 3 & if\ x < 8 \end{cases}$$

A. 73
B. 15
C. 60
D. 30
E. 89

5. 这个程序是做什么的？（分支结构）

以下哪个输入值在执行时会输出“ wrong ”？

```
input a, b
a = a ^ 2
b = b ^ 2
c = a + b
d = sqrt(c)
if d == int(d) then
     output a, b, c
else
     output "wrong"
end if
```

A. a=3, b=4
B. a=5, b=12
C. a=5, b=10
D. a=9, b=12
E. a=7, b=24

6. 这个程序是做什么的？（分支结构）

用 1 到 25 之间的每个数字（含首尾）依次运行以下程序，最终会输出多少个数字？

```
input num
if num % 3 == 0 && num % 4 == 0 then
     output num
end if
if num % 5 == 0 || num % 7 == 0 then
     output num
end if
```

A. 11
B. 10
C. 9
D. 8
E. 7